



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI ARUNACHALA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

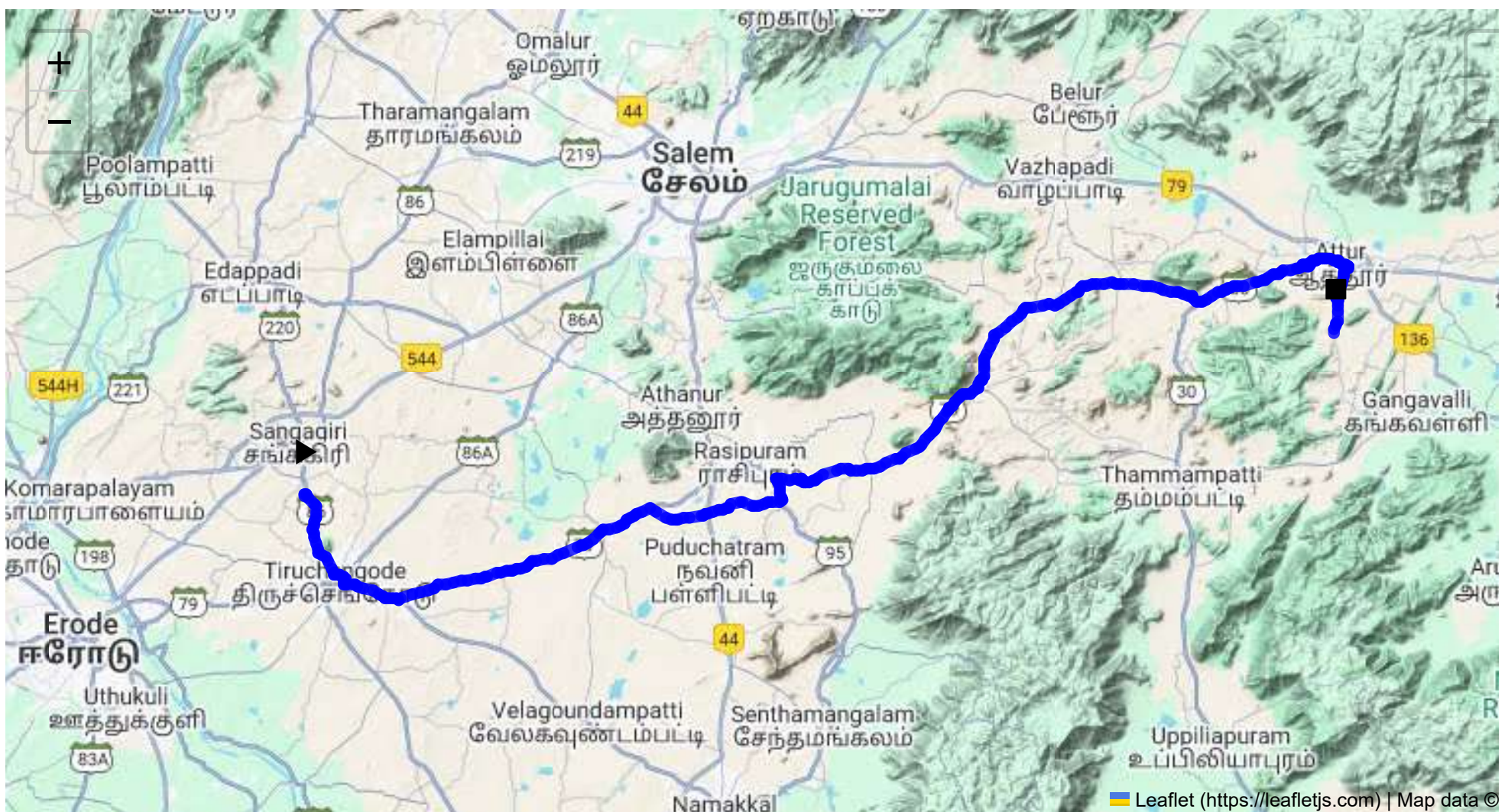
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 104.78 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (11.550599, 78.593239)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, Tamil Nadu to HH2V+67 Thulukkanur, Tamil Nadu covers approximately 105 kilometers. This path passes through several significant points: Sangagiri, Tiruchengode, Salem Main Road, santhapattai, Thokkavadi, Vinayagapuram, and Ambedkar Nagar. The route comprises both national highways and local roads with varying traffic densities.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The monsoon period from October to December can bring heavy rains, leading to potential flooding and road slippages, especially in rural locales. During the dry months, dust storms and high temperatures could affect visibility and vehicle performance.

3. Analysis of Traffic Patterns

- **Peak Hours:** The major town areas like Thiruchencode and Salem might experience congestion during morning and evening peak hours, typically from 8-10 AM and 5-8 PM.
- **Congestion-Prone Areas:** Key junctions around Salem Main Road and in town centers could lead to slow-moving traffic. Near markets and schools, expect heavy pedestrian and two-wheeler traffic.

4. Assessment of Road Quality and Infrastructure

- **National Highways:** Generally well-maintained, though periodic potholes and road work could occur, especially after the monsoon season.
- **Local Roads:** May have narrower lanes and uneven surfaces in rural segments like santhapattai and Attur.

5. Suggestions for Alternative Routes

- In case of road closure or severe traffic, consider using state highways connecting nearby villages as detour routes. However, these might be less developed and have lower capacity for heavy vehicles.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Local regulations require permits for transporting hazardous materials, particularly in high-density urban areas. There are restrictions on movement during festivals and peak hours in towns.

7. Overview of Historical Incidents

There have been sporadic incidents of road accidents involving heavy vehicles, primarily due to overtaking maneuvers and vehicular breakdowns. No major hazardous material spills were noted historically but caution is suggested near populous zones.

8. Environmental Considerations and Sensitive Areas

The route crosses agricultural regions where chemical spills could severely impact the environment. The vicinity to water bodies during monsoon could expand risks of contamination.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable on national highways; however, there may be signal dropouts in remote rural areas. Ensure alternate communication methods or satellite phones for uninterrupted connectivity.

10. Estimated Emergency Response Times

- **Urban Segments (near Salem and Attur):** Emergency services can typically respond within 20 to 30 minutes.
- **Rural Areas (e.g., Thokkavadi):** Response times could extend to 45 minutes or longer due to distance and accessibility issues.

11. Overall Summary of Risk Assessment

This route presents standard risks associated with rural and urban transit, including weather-related disruptions, traffic congestion, and potential road maintenance challenges. Heavy vehicle operators should maintain heightened alertness to road and weather conditions, adhere to local transportation laws, and prepare for communication interruptions. Emergency planning and alternative route charts are necessary for minimizing risks effectively. Overall, with due diligence and preparation, this route is navigable for heavy vehicles carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38382, 77.89897	30 KM/Hr
5	Turn	High	11.38188, 77.90257	15 KM/Hr
6	Turn	High	11.37830, 77.90160	15 KM/Hr
7	Turn	Medium	11.36754, 77.93919	30 KM/Hr
8	Turn	Medium	11.41361, 78.08073	30 KM/Hr
9	Blind Spot	Blind Spot	11.41453, 78.08062	10 KM/Hr
10	Turn	High	11.42956, 78.11590	15 KM/Hr
11	Turn	Medium	11.42185, 78.13013	30 KM/Hr
12	Turn	Medium	11.42450, 78.15278	30 KM/Hr
13	Turn	Medium	11.43432, 78.20586	30 KM/Hr
14	Turn	High	11.43542, 78.20754	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.45075, 78.20391	15 KM/Hr
16	Turn	High	11.45098, 78.20414	15 KM/Hr
17	Turn	High	11.44946, 78.22270	15 KM/Hr
18	Turn	High	11.44793, 78.22320	15 KM/Hr
19	Turn	Medium	11.44753, 78.22443	30 KM/Hr
20	Turn	Medium	11.45361, 78.23597	30 KM/Hr
21	Turn	Medium	11.49628, 78.32342	30 KM/Hr
22	Turn	Medium	11.50742, 78.33261	30 KM/Hr
23	Turn	Medium	11.50868, 78.33972	30 KM/Hr
24	Turn	Medium	11.51207, 78.34431	30 KM/Hr
25	Turn	Medium	11.51872, 78.34817	30 KM/Hr
26	Turn	Medium	11.54849, 78.35634	30 KM/Hr
27	Turn	Medium	11.56722, 78.37819	30 KM/Hr
28	Turn	Medium	11.58331, 78.44344	30 KM/Hr
29	Turn	Medium	11.57494, 78.49402	30 KM/Hr
30	Turn	Medium	11.57100, 78.49943	30 KM/Hr
31	Turn	Medium	11.57119, 78.50112	30 KM/Hr
32	Turn	Medium	11.57835, 78.51258	30 KM/Hr
33	Turn	Medium	11.58682, 78.54343	30 KM/Hr
34	Turn	Medium	11.58680, 78.54563	30 KM/Hr
35	Turn	Medium	11.59006, 78.55138	30 KM/Hr
36	Turn	Medium	11.59117, 78.55644	30 KM/Hr
37	Turn	Medium	11.59256, 78.55789	30 KM/Hr
38	Turn	High	11.59744, 78.57347	15 KM/Hr
39	Turn	High	11.59550, 78.57463	15 KM/Hr
40	Turn	High	11.59533, 78.57596	15 KM/Hr
41	Turn	High	11.59889, 78.57708	15 KM/Hr
42	Turn	Medium	11.60115, 78.58223	30 KM/Hr
43	Turn	High	11.59967, 78.59630	15 KM/Hr
44	Turn	Medium	11.59826, 78.59683	30 KM/Hr
45	Turn	High	11.59624, 78.60348	15 KM/Hr
46	Turn	Medium	11.59515, 78.60338	30 KM/Hr
47	Turn	High	11.59492, 78.60322	15 KM/Hr
48	Turn	High	11.59488, 78.60191	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
49	Turn	High	11.57661, 78.59570	15 KM/Hr
50	Turn	Medium	11.57646, 78.59596	30 KM/Hr
51	Turn	Medium	11.57559, 78.59622	30 KM/Hr
52	Turn	High	11.57406, 78.59547	15 KM/Hr
53	Turn	High	11.57318, 78.59613	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
10	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
11	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
12	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
14	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium
17	hospital	Sundaram Multi Speciality Hospital	11.600633, 78.5948143	30 km/h	Medium
18	clinic	RK Clinic	11.59969, 78.5941386	30 km/h	Medium
19	clinic	Attur ENT and Dental Care Centre	11.5991073, 78.5962503	30 km/h	Medium
20	hospital	Kalyani Memorial Hospital	11.599983, 78.596223	30 km/h	Medium
21	clinic	Mother's Ortho Clinic	11.6002438, 78.594053	30 km/h	Medium
22	hospital	Arun Hospital	11.599848, 78.596343	30 km/h	Medium
23	hospital	Chiranjeevi Hospital	11.5998283, 78.594309	30 km/h	Medium
24	clinic	Ramakrishna Nursing Home	11.5988349, 78.5988615	30 km/h	Medium
25	clinic	Sri Vinayaga Physiotherapy Clinic	11.598134, 78.597401	30 km/h	Medium
26	hospital	Sri Andavhar Nursing Home	11.6010351, 78.5974614	30 km/h	Medium
27	clinic	SS Clinic	11.5975084, 78.5979036	30 km/h	Medium
28	hospital	Gokulam Hospital	11.597356, 78.598403	30 km/h	Medium
29	clinic	Bawaa Clinic	11.597934, 78.597143	30 km/h	Medium
30	hospital	Kumar Eye Hospital	11.5978065, 78.5984006	30 km/h	Medium
31	hospital	N.S. Kumudham Hospital	11.59626, 78.599512	30 km/h	Medium
32	hospital	Rk Memorial Hospital	11.5975972, 78.5993824	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	hospital	Anugraha Hospital	11.5963566, 78.5941161	30 km/h	Medium
34	hospital	Sakthi Sivam Hospital	11.5959889, 78.5941723	30 km/h	Medium
35	hospital	Geeth Ragunath Hospital	11.5954889, 78.5955383	30 km/h	Medium
36	hospital	Panneerselvam Hospital	11.598067, 78.599939	30 km/h	Medium
37	hospital	Sri Dhanvantari Children Hospital	11.5991194, 78.5999856	30 km/h	Medium
38	hospital	Government Hospital Attur	11.59709, 78.601343	30 km/h	Medium
39	hospital	Abhi Eye Hospital	11.5980207, 78.6003698	30 km/h	Medium
40	hospital	Government Hospital	11.597079, 78.600619	30 km/h	Medium
41	clinic	Dr. Ramdoss Clinic	11.595518, 78.600865	30 km/h	Medium
42	hospital	Arul Gandhi Physiotherapy Hospital	11.5979315, 78.6003262	30 km/h	Medium
43	hospital	St. Joseph Hospital	11.5990918, 78.6003644	30 km/h	Medium
44	clinic	G.V. Homeo Clinic	11.596099, 78.600999	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
13	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
15	school	Government Boys Higher Secondary School	11.599149, 78.5888931	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	school	Government Boys Higher Secondary School	11.598743, 78.5885472	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38382, 77.89897



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38188, 77.90257



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37830, 77.90160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36754, 77.93919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41361, 78.08073



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41453, 78.08062



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42956, 78.11590



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42450, 78.15278



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43542, 78.20754



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45075, 78.20391



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45098, 78.20414



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44946, 78.22270



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44793, 78.22320



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44753, 78.22443



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45361, 78.23597



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49628, 78.32342



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50742, 78.33261



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50868, 78.33972



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51207, 78.34431



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51872, 78.34817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54849, 78.35634



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56722, 78.37819



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58331, 78.44344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57494, 78.49402



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57100, 78.49943



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57119, 78.50112



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57835, 78.51258



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58680, 78.54563



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59744, 78.57347



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59550, 78.57463



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59533, 78.57596



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59889, 78.57708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60115, 78.58223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59967, 78.59630



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59826, 78.59683



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59624, 78.60348



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59488, 78.60191
